

LESSON 306 (2.0GROUND)

Lesson Objective

During this ground lesson, the student will be introduced to navigation systems and the procedures used in the IFR environment.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

- Instrument Flying Handbook. ([FAA-H-8083-15B](#))
 - Chapter 09 - Navigation Systems.
- Aeronautical Information Manual. ([AIM](#))
 - Chapter 01 - Navigation Aids.
- FIT Warrior Maneuvers Manual.
 - Page 53-60.

Lesson Content

Navigation Instrument Interpretation

- *Use a graphic or simulator to assist with orientation of the appropriate systems aboard the training aircraft.*
- *This is a review to ensure that the student possesses an understanding of the equipment involved in intercepting and tracking.*
- CDI Orientation:
 - Review the parts of the instrument.
 - An annunciator built into the CDI as well as the G430 will determine whether the instrument shows GPS or VLOC data.
 - If proceeding inbound to a station, the instrument is set to a TO indication.
 - If proceeding outbound from a station, the instrument is set to a FROM indication.
 - If the course is set reciprocal to the direction of flight, reverse sensing results.
 - This instrument relays NO heading information – ONLY position information.
 - With a TO indication, radials are read at the bottom of the instrument with the CDI centered.
 - With a FROM indication, radials are read at the top of the instrument with the CDI centered.
 - Headings for intercepting may be read to the sides of the course index toward the CDI.
- HSI Orientation:
 - Review the basic parts of the instrument.
 - This instrument combines heading and navigation information, essentially representing a

primitive moving map.

- It is easy to visualize the aircraft's relationship to courses and waypoints.
- For VOR/GPS tracking, there is no reverse sensing.
- To avoid reverse sensing with localizers/back courses, always set the course to the front course.
- For the purposes of autopilot usage, when tracking TO the station, use a TO indication; when tracking FROM, use a FROM.
- With the CDI centered, the TO/FROM flag always points to the station.
- Inbound Courses are read at the arrowhead with a TO indication.
- Radials are read at the tail of the arrow with a TO indication OR at the head of the arrow with a FROM indication.

- **RMI Orientation:**

- Keep in mind that for glass operators this instrument is combined with the HSI.
- This instrument combines heading and magnetic bearing information.
- The bearing pointer always points to the navaid/waypoint.
- The advantage of this instrument is that one does not have to center a CDI to determine the position information.
- Course and Radial information are read instantaneously.
- Inbound Courses and Bearings To are at the head of the arrow.
- Radials and Bearings From are read at the tail.
- Station passage is defined by the reversing and settling at the 180° position.

VOR Intercepting and Tracking

- *Use a graphic or simulator to assist with orientation of the appropriate systems aboard the training aircraft.*
- **VOR Signal Characteristics:**
 - Radials determine where the aircraft is FROM the station.
 - (Inbound) Courses determine the track that will take the aircraft TO the station.
 - Radials are angular to each other – (Lateral Distance per Radial = NM from Station ÷ 60).
 - This is why the CDI is more sensitive close to the station.
 - When overflying VORs, within approximately 1 nm, due to the sensitivity, hold heading until station passage.
 - Station passage is defined by the complete, positive reversal of the TO/FROM indicator.
- **Intercepting Outside of 5NM (Warrior Maneuvers Manual):**
 - This procedure is for intercepting a radial other than the one the aircraft is presently on.
 - *Any time a VOR is used, tune and identify the station.*
 - Verify that the HSI/CDI is set up to the proper radio (VLOC1/VLOC2).
 - Determine the difference between the current and the desired radial and double.
 - Rotate the OBS to the desired radial/course.
 - Apply the intercept angle to the desired course to find the intercept heading.
 - Turn the aircraft in the quickest path to the intercept heading.

Instrument Rating Course

- Once the CDI moves from full-scale deflection, call “needle alive”.
- The distance from the station and groundspeed determines how early to turn on course.
- Show the student that a practical application of intercepting radials is intercepting Victor Airways.
- Intercepting Within 5NM (Warrior Maneuvers Manual):
 - This situation is often encountered shortly after departure.
 - This is also encountered after crossing VORs as initial or final approach fixes.
 - A 30° - 45° angle is used to prevent overshooting the sensitive courses.
 - Whenever intercepting problems are given, crosscheck DME.
- Tracking (Warrior Maneuvers Manual):
 - Courses are maintained as accurately as possible to maintain situational awareness and obstruction clearance.
 - To track inbound to a station, center the CDI with a TO indication.
 - To track outbound from a station, center the CDI with a FROM indication.
 - Turn to the course indicated by the index and fly that heading.
 - If the CDI moves off center, make an Initial correction of 20 degrees in the direction of the CDI.
 - When the CDI centers again, reduce the correction by half.
 - Keep the correction in (this helps fix wind drift) and fix any CDI deflection by small heading changes.
 - In advanced glass systems, make use of the ground track vector/bug.
- Common Errors:
 - Forgetting to tune and ID.
 - Applying intercept angle to headings or current radials.
 - Not setting up HSI/CDI properly to the correct navaid or source.
 - Making grossly large corrections, especially close to the station.
 - Homing – Often, students will note a deflection and center the CDI to a new course to correct for drift.
 - Losing track of terms such as radial, course, inbound, and outbound.
 - Not utilizing the TO/FROM flag for situational awareness.
 - Not flying the shortest path to the intercept heading.
 - Undershooting and overshooting courses.

DME Arc Intercepting and Tracking

- *DME arcs are courses used in departures, joining approach courses, and rarely, final approach courses themselves.*
- *Either draw out the procedures or demonstrate them in a simulator.*
- *It is a good idea to show the student an example of how published arcs work.*
- Intercepting from a Radial (Warrior Maneuvers Manual):
 - Tune and identify the VOR.
 - Stress to note the compass direction of the arc.

- As early as practical, calculate the intercept heading based on the current radial.
- The no-wind intercept is 100° to the present radial (Advance OBS 10° and add 90° in direction of the arc).
- Join the arc at 0.5 nm prior to the arc (depending on wind conditions) – immediately advance OBS 10° .
- If undershooting the arc, roll out early; If overshooting, continue to turn to intercept the arc.
- Intercepting from a Radar Vector (Warrior Maneuvers Manual):
 - Tune and identify the VOR.
 - Stress to note the compass direction of the arc.
 - At 1.5 nm prior to the arc, calculate the intercept heading based on the current radial.
 - The no-wind intercept is 100° to the present radial (Advance OBS 10° and add 90° in direction of the arc).
 - Join the arc at 0.5 nm prior to the arc (depending on wind and how many degrees the turn requires) – immediately advance OBS 10° .
 - If undershooting the arc, roll out early; If overshooting, continue to turn to intercept the arc.
 - The intercept angle is generally less than 90° , so the turn may need to be delayed preventing undershooting.
- Intercepting from a GPS Fix (Warrior Maneuvers Manual):
 - The instrument approach desired will need to be loaded from an IAF transition that begins an arc segment.
 - GPS will need to be active on the HSI to navigate to the IAF.
 - Upon crossing the IAF, GPS can be used to track the arc.
 - The HSI will be tangential to the arc – it will update as the course is flown.
 - The CDI will indicate deviation from the arc, replacing the numerical course information as before.
 - Prior to joining a VLOC course as part of the procedure, the CDI must be on VLOC, not GPS.
- Tracking DME Arcs - Turn 10/Twist 10 Method:
 - By default, the HSI/CDI is used to track the arc
 - After intercepting every 10th radial, Turn 10, Twist 10.
 - If ground track information is shown on the HSI, tracking 90° to the set radial will hold the arc within tolerances.
 - If there is drift, DME will change – Use displayed wind information to prevent this from occurring.
 - Inside the arc means that the DME is less than the arc value – For small deviations, hold heading. Else, turn 10° .
 - Outside the arc means that the DME is greater than the arc value – Turn 10° at a time to correct.
 - No matter the wind correction, always twist the OBS.

- Tracking DME Arcs - Bearing Pointer Method:
 - Instead of Turn 10 Twist 10, the pilot turns 10° after allowing the pointer to traverse 10° on the heading indicator.
 - In no wind, one can use the wingtip reference marks – The head falls 5° ahead to 5° behind (opposite for tail).
 - Determining radials is just done by noting the tail of the arrow – there is nothing to twist.
 - If ground track information is shown on the HSI, tracking 90° to the pointer will hold the arc within tolerances.
 - If there is drift, change heading 10° at a time to reacquire the arc.
- Common Errors:
 - Forgetting to tune and ID DME and courses being arced to.
 - Not setting up navigation sources properly before intercepting the arc.
 - Miscalculating the intercept heading.
 - Forgetting to twist the OBS.
 - Overcorrecting for wind drift.
 - Not turning to intercept the arc at the appropriate distance, or not adjusting for ground speed or intercept angle.
 - Not utilizing the TO/FROM flag for situational awareness.
 - Overflying courses to be intercepted.

Localizer Intercepting and Tracking

- Localizer Orientation:
 - The service volume is angular – 35° up to 10 nm and 10° up to 18 nm – Here are the area's best to tune and identify.
 - Localizers are always identified with two dots in their Morse code preceding their identifier (I-MLB).
 - Explain that the signal is directional – different courses cannot be selected and bearing pointers will NOT work.
 - Review the difference between the front course and the back course.
 - Back courses are more sensitive because the antenna is closer to the aircraft tracking them.
 - To avoid reverse sensing, set the HSI to the front course.
 - Conventional receivers should be set to the inbound course for situational awareness.
 - On average, the localizer is four times as sensitive as a VOR and is more sensitive closer to the antenna.
 - Inbound on a localizer or back course means tracking toward the landing runway.
 - Show the student how to find localizer information in the TERPS in order to set them up.
- Intercepting a Localizer:
 - Tune and identify the localizer when within service volume.
 - Twist the front course.
 - Determine your current location in relation to the localizer course.

- Lead the intercept by the amount required – this will depend on the distance from the antenna.
- It is advisable to begin a turn at or shortly after the CDI moves from full-scale deflection – Call “needle alive”.
- ATC often assigns an intercept angle of 20° - 30°.
- DME Arcs are common ways of joining localizers.
- VOR radials (or other navaid signals) called ‘feeder routes’ allow one to join a localizer.
- A procedure turn often places the aircraft inbound on a localizer to allow the execution of an approach.
- Tracking a Localizer:
 - A localizer is much more sensitive, so corrections for deviations should be smaller.
 - Discuss when conventional receivers display reverse sensing (outbound on the front course, inbound on the back course).
 - If the CDI is off center, make an Initial correction of no more than 10 degrees in the direction of the deflection.
 - When the CDI centers again, reduce the correction by half; this will correct for wind drift.
 - Make corrections no larger than 2-3 degrees at a time.
 - All turns after passing the final approach fix should be half standard rate.
- Common Errors:
 - Forgetting to tune and ID the station.
 - Chasing needles.
 - Turning too late.
 - Insufficient reaction to deviations.
 - Disorientation due to improper setup and reliance on minimum equipment.

NDB Intercepting and Tracking

- *It is possible to simulate ADF navigation in glass systems by utilizing bearing pointers and GPS waypoints/VORs*
- *It is advisable to show students who are likely to fly outside of the United States*
- Intercepting NDB Bearings - (Warrior Operations Manual):
 - Tune and continuously identify the station.
 - Ensure that the radio is in ADF mode.
 - Determine your location to the NDB.
 - Fly parallel to the desired bearing – reset the moveable card.
 - Determine the amount of deflection from zero and double it for intercept angle.
 - Apply the intercept angle to the present heading in the direction of the head of the needle (inbound or outbound).
 - Turn the aircraft to the intercept heading – reset the moveable card.
 - Maintain heading until the needle deflects the number of degrees from the zero position as the intercept angle (minus lead).
 - An alternative to the above is to wait until the head or tail of the needle matches the

desired magnetic bearing (minus lead).

- Turn to the desired bearing and correct for wind.
- **NDB Tracking - (Warrior Operations Manual):**
 - Firstly, explain the difference between tracking and homing.
 - To track inbound, turn the aircraft to produce zero relative bearings.
 - Maintain the heading until drift is detected.
 - Correct 20° initially.
 - After reestablishing the aircraft on course, use a 10° correction angle.
 - For future deviations, repeat the previous steps - (Bracketing).
- **Common Errors:**
 - Failing to tune and identify.
 - Not setting the heading indicator to the compass.
 - Neglecting the use of moveable cards.
 - Disregarding any RMI warning indications.
 - HOMING.
 - Not following the proper intercept procedure – often forgetting to parallel.
 - Chasing the needle close to the station/waypoint.

GPS Intercepting and Tracking

- **Direct-To Nav**
 - All one needs to do is press DIRECT
 - Type the destination and press ENTER twice
 - To cancel Direct-To NAV, press DIRECT
 - Next press MENU, and select CANCEL DIRECT-TO NAV, and press ENTER
- **Flight Plans**
 - Press FLP
 - Use the FMS to activate the cursor and type desired waypoints
 - To change a LEG SEGMENT, highlight the SECOND WAYPOINT of the segment
 - Press MENU, select ACTIVATE LEG, and press ENTER
- **OBS Mode**
 - This mode can be likened to “turning a GPS waypoint into a VOR” – multiple bearings can be selected.
 - On a loaded procedure/flight plan, this mode suspends waypoint cycling (CRUCIAL for GPS holds).
 - To engage, press the OBS pushbutton on the G430, OR, on the G1000, press the OBS softkey on the PFD.
 - VOR intercepting and tracking procedures can then be used.
 - Note that the CDI does not become more sensitive toward the waypoint.
 - Pressing OBS again will turn the mode off and resume normal waypoint cycling.
- **Common Errors**
 - Lack of knowledge of the system, resulting in mashing random buttons.

- Mismanaging OBS mode when it is needed.
- Not monitoring the active waypoints and legs in the system.
- Forgetting to verify GPS modes.

Autopilot

- *Always verify that the correct navigation source is selected on the HSI.*
- *Monitor the actions of the autopilot – if it is not performing as expected, disconnect.*
 - The STEC-55X will intercept a course at a default 45° angle when NAV is pressed.
 - The combination HDG + NAV prompts the autopilot to capture a selected course from the bugged heading.
 - Within 15% scale deflection, CAP mode will limit sensitivity to closure to complete the intercept.
 - 15 seconds after capture, CAP SOFT will show, and the turn rate is limited to 45%, and sensitivity to closure is reduced.
 - 30 seconds after capture, crosswind corrections are made.
 - 75 seconds after capture, SOFT will annunciate, turn rate is limited to 15%, and sensitivity to closure is further reduced.
 - APR will establish CAP SOFT authority to keep the aircraft on course.
 - REV will track the opposite of what the course index is set to – This is very useful for tracking back courses.
 - When VLOC signals, NAV mode allows up to 50% deflection to prevent S-Turns when passing VORs
 - After 60 seconds in this condition, CAP SOFT is reestablished to intercept the course.
 - To track GPS signals, press NAV twice to engage GPSS – courses may be intercepted like before but with 2 presses of NAV.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 307.
- Assign Homework:
 - Instrument Flying Handbook:
 - Chapter 09 - Navigation Systems.
 - FIT procedures for VOR Intercepting and Tracking.
 - FIT procedures for Intercepting and Tracking DME Arcs from radials and radar vectors.

Completion Standards

Through oral quizzing, the student should demonstrate a basic understanding of each of the navigation aids and concepts for using them.